

Adjoint-Based Thermal-Security Robustness Radii for AC Power Systems:

A Post-Dispatch Screening and Diagnostic Certificate Under Response-Constrained Injection Uncertainty

Egor Grishin

Center for Energy Science and Technology, Moscow, Russia
mkgrgor@gmail.com

Revision R1

Abstract

This paper develops a constrained dual-norm framework for first-order robustness radii of power-system operating points and specializes it to AC thermal security. The radius is the largest metric ball of admissible injection perturbations, restricted by an uncertainty-response model, that remains feasible for all monitored affine constraint surrogates. For AC networks, two-ended apparent-power constraints are mapped to injection space through sparse transposed-Jacobian solves, avoiding dense sensitivity matrices; zero-flow line ends, where the apparent-power magnitude is nondifferentiable, receive an exact first-order operator-norm radius, so the certificate covers every monitored constraint. The framework covers Euclidean, covariance-weighted, and sigma radii, yields worst-case perturbation directions, and extends to steady-state security-distance concepts with explicit response constraints. We prove exact affine-model maximality, DC slack-reference invariance, and metric-sigma equivalence. The contribution is positioned strictly as a *post-dispatch screening and diagnostic layer*: it evaluates a fixed operating point produced by any dispatch process and certifies the affine surrogate only. Verification is correspondingly thorough: the assembled adjoint derivative now agrees with centered finite differences to the power-flow-tolerance floor (relative errors decreasing as $O(\varepsilon^2)$ from 10^{-6} to 10^{-10}); line-wise analytical flow standard deviations match nonlinear Monte Carlo within $\pm 2\%$ (median ratio 1.004), making the sigma radius a calibrated first-order index in the tested regime; multi-scale nonlinear replay of worst-case directions quantifies the mild nonconservatism of the affine radius (empirical crossing at 0.90–1.00 of the certified radius); and danger-oriented radius rankings outperform loading- and headroom-only indicators with scenario-bootstrap confidence intervals that exclude zero.

Keywords: power-system security; security distance; robustness radius; post-dispatch screening; AC power flow; adjoint sensitivity; thermal limits; uncertainty quantification.

1 Introduction

Power-system operators need fast, interpretable security margins around an operating point that has already been selected — by an OPF, a security-constrained OPF, a market process, or manual procedures. The question this paper answers is strictly *post-dispatch*: given a fixed AC operating point, how large is the admissible set of continuous injection perturbations before some monitored thermal constraint is reached, which constraint is it, and which injection pattern gets there first?

Finite-contingency SCOPF does not answer this continuous, high-dimensional question, and fast operational metrics — loading ratio, thermal headroom, PTDF sensitivities, LODF screening [1] — measure either margin or sensitivity but not a constrained multidimensional distance to violation. Robust and chance-constrained OPF internalize uncertainty during dispatch [2–6], but they require a chosen uncertainty model and a new dispatch solve, and they do not evaluate an arbitrary existing operating point.

The idea of measuring security as a *distance* from an operating point to a security-region boundary is established: steady-state security distance (SSSD) combines operating margin with boundary sensitivity and has been used for N–1 assessment [7–9]. This paper is a constrained-geometry, AC-adjoint member of that family, not the first distance-to-boundary construction; the precise positioning is given in Section 2 and Table 1. Relative to SSSD, the radius developed here (i) restricts perturbations to an explicit uncertainty-*response* subspace $\{\Delta u : A\Delta u = 0\}$ or response map $\Delta u = T\omega$, so balancing, participation, and reactive-support conventions are part of the geometry; (ii) supports general metrics $M \succ 0$, recovering covariance-weighted (sigma) radii as inverse-covariance metric radii; (iii) evaluates *two-ended AC apparent-power* constraints through sparse transposed-Jacobian (adjoint) solves, without dense sensitivity matrices; and (iv) returns closed-form worst-case directions, including an operator-norm treatment of zero-flow ends where the scalar gradient does not exist.

The contribution is a unified certificate model and a verified algorithmic framework:

1. a constrained dual-norm radius formula for affine engineering constraints, with exact affine-model maximality, worst-case directions, DC slack invariance, and metric-sigma equivalence;
2. DC and AC thermal-security specializations with two-ended apparent-power constraints, a reactive-limit-aware active set, and a first-order operator-norm certificate for zero-flow ends, so that *every* monitored constraint receives a radius;
3. a sparse AC adjoint implementation whose derivative is verified to the power-flow-tolerance floor by step-sweep centered finite differences and per-solve adjoint residuals;
4. a screening-oriented validation methodology: line-wise sigma calibration against nonlinear Monte Carlo, multi-scale worst-case replay quantifying affine nonconservatism, scenario-bootstrap ranking statistics with paired confidence intervals and precision/recall, and a physically realizable participation-factor response study.

Scope statement, stated once and used throughout: the certificate is *exact for the affine surrogate* and is a *local first-order screening indicator* for the nonlinear AC system. It is not a nonlinear feasibility certificate, and this paper does not propose a dispatch optimization method; radius-aware dispatch is left to future work precisely because a fair evaluation requires matched-cost baselines and a full SCOPF comparator.

2 Literature Review

Steady-state security distance SSSD defines the distance from an operating point to a steady-state security-region boundary and uses it for assessment and ranking [7, 8]; the N–1 extension screens contingencies through the same distance [9]. The present radius differs in the uncertainty geometry (equality-constrained ellipsoids and covariance metrics instead of a fixed Euclidean injection norm), the constraint model (two-ended AC apparent-power limits handled by adjoint solves

Table 1: Positioning relative to the steady-state security-distance and robustness-diagnostics literature. SSSD = steady-state security distance [7–9]; CIA = convex inner approximations of steady-state security regions [10]; UD = unified AC operating-point robustness diagnostics [11].

	SSSD	CIA	UD	This work
Uncertainty space	bus injections, fixed directions/boundary distance	full nonlinear feasible set	injections + topology, regime changes	response-constrained injections $\{\Delta u : A\Delta u = 0\}$
Metric	Euclidean distance to boundary	set inclusion (no scalar metric)	sensitivity-normalized margins	any $M \succ 0$: Euclidean, per-unit, covariance (Σ^{-1})
Constraints	steady-state security region boundary	nonlinear AC feasibility	multiple diagnostic families	two-ended AC apparent-power thermal limits
Network model	DC / PV-curve based boundaries	nonlinear AC with bounded remainders	AC sensitivities around solved PF	AC first-order (reduced Jacobian, lossless series model)
Contingency treatment	N–1 via boundary recomputation	base topology	topological uncertainty classes	DC LODF screening layer (diagnostic only)
Nonlinear guarantee	none (boundary linearization)	yes (inner approximation)	none (local)	none beyond affine surrogate; empirically quantified crossing ratios
Worst direction	not explicit	not applicable	partially (per diagnostic)	closed form $K_M(A)h/\sqrt{h^\top K_M(A)h}$; operator-norm at $ S_0 =0$
Complexity	boundary construction per scenario	SDP/convex program	per-diagnostic linear algebra	one sparse LU + $2m$ triangular solves (chunked)

instead of DC or boundary-wise linearizations), the explicit worst-direction recovery, and the computational profile (one sparse LU plus $2m$ triangular solves). Table 1 gives the side-by-side comparison requested by the steady-state security-distance literature.

Nonlinear inner approximations Convex inner approximations construct genuinely nonlinear feasibility certificates by bounding nonlinear terms [10]. They provide stronger guarantees at higher computational cost; the radius developed here trades the nonlinear guarantee for closed-form evaluation and scalable adjoint linear algebra, and Section 6 quantifies empirically how far the affine radius sits from the nonlinear crossing distance.

Post-solution diagnostics Recent unified diagnostics quantify AC operating-point robustness under injection and topological uncertainty with regime changes [11]. That framework spans a broader diagnostic family (including topology and regime switching); the present work is narrower and more formulaic: for thermal constraints in a fixed affine regime it provides a closed-form constrained ellipsoidal radius, an exact worst-case direction, and a two-ended sparse adjoint implementation with a machine-verifiable derivative. The two approaches are complementary: the radius could serve as one diagnostic inside such a unified suite.

Robust and chance-constrained OPF Robust optimization provides feasibility guarantees over uncertainty sets [2, 3]; chance-constrained OPF enforces probabilistic security levels [4–6]; robust convex restrictions target nonlinear AC feasibility [12]. These are dispatch-side tools; the radius is an evaluation-side tool that measures the robustness such methods achieve, for whatever operating point they produce.

Table 2: Notation and units.

Symbol	Units	Meaning
$\mathcal{N}, \mathcal{L}$	–	bus set ($ \mathcal{N} = n$), monitored line set ($ \mathcal{L} = m$)
ω	MW, MVA	primitive uncertainty (forecast errors)
T	–	response map (participation, balancing, coordinate reduction)
Δu	MW, MVA	response-coordinate perturbation, $\Delta u = T\omega$
A, C	–	admissibility (equality) matrices in response / primitive space
M, W	$(\text{MW})^{-2}$	metric matrices in response / primitive space
Σ	$(\text{MW})^2, (\text{MVA})^2$	injection covariance; $M = \Sigma^{-1}$ gives sigma radii
$K_M(A)$	$(\text{MW})^2$	inverse metric restricted to $\ker A$
x	rad, p.u.	reduced AC state $[\theta_{\mathcal{N} \setminus s}; V_{\mathcal{P} \cup \mathcal{Q}}]$
J	MW/rad, MW/p.u.	reduced AC power-flow Jacobian at the base point
$b_{\ell,e}$	MVA/rad, MVA/p.u.	state gradient of $s_{\ell,e}$
$h_{\ell,e}$	MVA/MW	injection-space sensitivity, $J^\top h_{\ell,e} = b_{\ell,e}$
$s_{\ell,e}, c_\ell$	MVA	line-end apparent power; thermal limit
$M_{\ell,e} = c_\ell - s_{\ell,e}^0$	MVA	thermal margin at end (ℓ, e)
r_ℓ^{DC}	MW	DC radius (balanced Euclidean MW injection norm)
$r_{\ell,e}^{AC,M}$	metric-dep.	AC end radius: MW-scale for Euclidean M ; dimensionless standard deviations for $M = \Sigma^{-1}$
β_j	–	standardized affine margin (sigma radius) of constraint j
$\sigma_{\max}$	MVA/MW	operator norm of the projected 2D (P, Q) response at a zero-flow end

Reliability indices and sensitivity screening The sigma radius is structurally a first-order reliability index [13] specialized to affine thermal surrogates with response-constrained Gaussian geometry; Section 6 reports the calibration evidence that justifies (and bounds) this interpretation. PTDF/LODF and loading-based screening [1] remain the fast baselines against which the radius is evaluated.

3 Problem Formulation

Table 2 collects notation and units.

General affine-radius model Consider an operating point with monitored affine surrogate constraints

$$y_j^0 + a_j^\top \Delta u \leq c_j, \quad j \in \mathcal{J}, \quad (1)$$

where Δu is the local response coordinate. In primitive space, uncertainty ω may pass through a response map $\Delta u = T\omega$ representing balancing participation, storage response, reactive-control assumptions, or coordinate reduction. If $\Omega_W(1) = \{\omega : C\omega = 0, \omega^\top W\omega \leq 1\}$, the support denominator of sensitivity h is $\sigma(h) = \sqrt{h^\top T K_W(C) T^\top h}$ with $W \succ 0$, C full row rank, and

$$K_W(C) = W^{-1} - W^{-1} C^\top (C W^{-1} C^\top)^{-1} C W^{-1}. \quad (2)$$

Alternatively, work directly in response space with $\mathcal{B} = \{\Delta u \in \mathbb{R}^d : A\Delta u = 0\}$, a metric $M \succ 0$, and the unit set $\mathcal{U}_M(1) = \{\Delta u : A\Delta u = 0, \Delta u^\top M \Delta u \leq 1\}$. The two descriptions agree when M is the metric induced by (T, W, C) on $\text{range}(T)$. For one-sided constraints the margin is $m_j = c_j - y_j^0$; a two-sided constraint contributes its two one-sided forms.

Definition 1 (Linearized robustness radius). *$r \geq 0$ is a linearized robustness-radius certificate if every $\Delta u \in r\mathcal{U}_M(1)$ satisfies all monitored affine constraints. The global radius r^* is the largest such r when all base margins are nonnegative. If any margin is negative, the operating point is reported as base-infeasible for that constraint rather than assigned a certified radius.*

Power-system specialization Let $\mathcal{N}$ ($|\mathcal{N}| = n$) be the buses and $\mathcal{L}$ ($|\mathcal{L}| = m$) the monitored lines with symmetric thermal limits $c_\ell > 0$. DC certificates monitor active-power flows f_ℓ (MW) under the balanced subspace $\mathcal{B}_P = \{\Delta p \in \mathbb{R}^n : \mathbf{1}^\top \Delta p = 0\}$. AC certificates monitor from- and to-end apparent-power magnitudes $s_{\ell,e}$ (MVA), $e \in \{f, t\}$, with the response coordinate $\Delta u = (\Delta P_{\mathcal{I}_P}, \Delta Q_{\mathcal{I}_Q})$ on the retained reduced mismatch coordinates. The admissibility matrix A encodes the chosen uncertainty-response convention, not a physical law: examples include net-zero active balance, blockwise active/reactive net-zero forecast error, or a participation-factor response map (Section 6 evaluates one with finite generator headroom). The radius is exact for the selected affine surrogate; for nonlinear AC power flow it is a local first-order indicator, and nonlinear replay is an independent validation layer, not part of the proof.

DC radius With DC sensitivities $\Delta f = H\Delta p$, headroom $M_\ell^{DC} = c_\ell - |f_\ell^0|$, and the ones-complement projection $\bar{g}_\ell = g_\ell - (\mathbf{1}^\top g_\ell/n)\mathbf{1}$, the line radius is $r_\ell^{DC} = M_\ell^{DC}/\|\bar{g}_\ell\|_2$ (finite case), with infeasible/inactive classifications as in the implementation.

AC radius Let $x = [\theta_{\mathcal{N}\setminus s}; V_{\mathcal{P}\mathcal{Q}}]$ be the reduced state and $J = \partial(P, Q)/\partial x$ the reduced Jacobian at the base point. *Active-set consistency*: the power flow is solved with reactive limits enforced; a bus whose every in-service voltage-controlling generator is saturated at a Q limit is effectively PQ in the converged solution and is linearized as PQ (its voltage variable and reactive equation enter x and J). This choice is validated to the power-flow-tolerance floor in Section 6. For $s_{\ell,e}^0 > 0$,

$$b_{\ell,e} = \frac{P_{\ell,e}^0 \nabla_x P_{\ell,e} + Q_{\ell,e}^0 \nabla_x Q_{\ell,e}}{s_{\ell,e}^0}, \quad J^\top h_{\ell,e} = b_{\ell,e}, \quad (3)$$

and the end radius is $r_{\ell,e}^{AC,M} = M_{\ell,e}^{AC}/\sigma_{\mathcal{U}_M(1)}(h_{\ell,e})$ with $M_{\ell,e}^{AC} = c_\ell - s_{\ell,e}^0$; the line takes the binding end and the system takes the minimum line.

Zero-flow ends At $s_{\ell,e}^0 = 0$ the magnitude has no scalar gradient, but the first-order model is still exact: $s_{\ell,e}(\Delta u) = \|(\nabla P^\top \Delta u, \nabla Q^\top \Delta u)\|_2 + O(\|\Delta u\|^2)$. With $H_{\ell,e} = [h_{\ell,e}^P; h_{\ell,e}^Q]$ obtained from two adjoint solves and projected onto the admissible subspace, the exact first-order radius is

$$r_{\ell,e} = \frac{c_\ell}{\sigma_{\max}(H_{\ell,e} \text{Proj}_{\ker A})}, \quad (4)$$

a 2×2 eigenvalue computation per end. The nondifferentiability threshold is scale-aware, $\max(10^{-9}, 10^{-9}c_\ell)$ MVA, and Section 6 reports its sensitivity. This closes the gap of the original submission, where zero-flow ends were excluded and large cases were only partially certified.

4 Method

Constrained support function and exact radius With $K_M(A) = M^{-1} - M^{-1}A^\top(AM^{-1}A^\top)^{-1}AM^{-1}$, the support function of the constrained unit ball is $\sigma_{\mathcal{U}_M(1)}(h) = \sqrt{h^\top K_M(A)h}$, the worst-case unit perturbation is $K_M(A)h/\sqrt{h^\top K_M(A)h}$, and:

Theorem 1 (Exact affine-model radius). *For one-sided constraints with nonnegative margins m_j , $r^* = \min_j m_j/\sqrt{a_j^\top K_M(A)a_j}$ is the largest radius for which every $\Delta u \in r^* \mathcal{U}_M(1)$ satisfies all affine constraints.*

The proof is the standard support-function argument for an ellipsoid intersected with a subspace, specialized to thermal surrogates; following the convex-geometry literature we present it as a unifying engineering specialization rather than a new theoretical result. Slack-reference invariance of the balanced DC radius and the metric-sigma equivalence $\beta_j = m_j / \sqrt{h_j^\top K_{\Sigma^{-1}}(A) h_j}$ (the conditional standard deviation of the affine response under $A\Delta u = 0$) carry over from the original submission unchanged; for a diagonal covariance and a single balance equation the projection is variance-weighted mean removal. We emphasize the distinction, raised during review, between *conditioning* a Gaussian vector on $A\Delta u = 0$ and *applying a physical response map* $T\omega$: the two coincide only when the conditional covariance equals $T\Sigma_\omega T^\top$, and every experiment below states which mechanism it uses.

Sparse adjoint algorithm and verification hooks The AC computation factorizes the reduced Jacobian once (sparse LU) and solves one transposed system per line end, in chunks; zero-flow ends add two solves. Two verification quantities are computed *inside* the certificate: the adjoint residual $\|J^\top h - b\|_\infty / \max(1, \|b\|_\infty)$ per chunk (reported with every run), and the step-sweep centered finite-difference protocol of Section 5 (reported with every case study). Peak adjoint right-hand-side storage is $O(n_x q)$ at chunk size q .

DC N-1 screening layer For DC contingency screening, LODF-updated sensitivities give post-contingency radii as a fast diagnostic layer. This is screening only: it is not a substitute for AC contingency simulation, and no dispatch procedure is built on it in this paper.

5 Verification and Evaluation Protocol

Benchmarks PGLib-OPF networks [14] parsed via MATPOWER-format inputs [15] into pandapower [16]; hourly demand profiles derived from UnitCommitment.jl instances [17] were used. The certificate model is the series-only lossless surrogate; the verification power flow applies the SAME policy, and a regression test asserts equality of the certificate admittance model and the verification solver’s internal admittance matrix, so the affine model and the replay model cannot drift apart silently.

Derivative verification For each case, the ten tightest differentiable binding ends are replayed with centered finite differences along the balanced worst-case direction at $\varepsilon \in \{1, 10^{-1}, 10^{-2}, 10^{-3}\}$ MVA. Acceptance requires the relative error to decrease with ε until the power-flow-tolerance floor; a plateau is treated as a model defect, not a diagnostic curiosity. The original submission’s plateau at the 10^{-2} level was traced to two implementation defects — an ext-grid-inconsistent slack index that shifted every active-power sensitivity entry by one bus position on affected cases, and an unmodeled reactive-limit active set — both fixed and regression-tested.

Statistical protocol Monte Carlo experiments use fixed seeds, report attempted/converged sample counts and PF failures, and quantify ranking uncertainty by *scenario-level* bootstrap (resampling Monte Carlo scenarios, which preserves the cross-line correlation structure), with paired bootstrap intervals for score differences and precision/recall at k for top-risk identification. Timing rows are mean $\pm$ std over 7 repetitions with measured peak resident memory.

Table 3: Adjoint verification after the implementation corrections: centered finite differences along the balanced worst-case direction of the ten tightest differentiable line ends per case. The relative error now decreases as $O(\varepsilon^2)$ down to the power-flow-tolerance floor, in contrast to the non-convergent 10^{-2} -level plateau of the original submission, which was traced to a slack-index defect and an unmodeled reactive-limit active set. Certificate time is mean of 7 repetitions (adjoint + norms, excluding AC PF).

Case	Ends	Med. rel. ($\varepsilon=1$)	Med. rel. ($\varepsilon=10^{-2}$)	Med. rel. ($\varepsilon=10^{-3}$)	Max rel. ($\varepsilon=10^{-3}$)	Adj. resid.	Cert. (s)
case14_ieee9		2.33×10^{-6}	1.98×10^{-10}	1.19×10^{-10}	3.12×10^{-10}	4.06×10^{-16}	0.0305
case30_ieee9		6.17×10^{-6}	6.16×10^{-10}	5.36×10^{-11}	6.38×10^{-10}	4.69×10^{-16}	0.0538
case118_ieee9		4.11×10^{-7}	3.08×10^{-11}	1.73×10^{-10}	5.51×10^{-10}	4.36×10^{-16}	0.5022
case200_activ		6.62×10^{-9}	7.46×10^{-11}	4.07×10^{-10}	1.68×10^{-9}	1.69×10^{-15}	0.4795

Table 4: Regenerated paired DC and AC all-constraint radii (ext-grid-consistent slack, Q-limit-aware active set, operator-norm zero-flow ends; AC certificate in the balanced two-block $[P; Q]$ Euclidean norm, AC-FPF base points).

Case	$r_{\star}^{DC}$	$r_{\star}^{AC}$	AC/DC	ND ends
case5_pjm	112.83	110.33	0.978	0
case14_ieee	121.42	118.88	0.979	0
case24_ieee_rts	7.90	4.20	0.532	0
case30_ieee	12.21	8.32	0.681	0
case57_ieee	27.52	23.54	0.855	0
case73_ieee_rts	25.70	11.64	0.453	0
case118_ieee	5.02	5.80	1.156	0
case200_activ	41.01	38.58	0.941	0

Reproducibility All experiments in this revision regenerate from one commit of the public repository via committed scripts (`experiments/revision_r1/scripts/`), with JSON artifacts committed alongside (`experiments/revision_r1/results/`). The revision experiments ran on a Linux container (48-thread Intel Xeon class, Python 3.11, pandapower 2.14, SciPy 1.13, NumPy 1.26); absolute timings therefore differ from the original submission’s hardware, and only relative structure should be compared.

6 Results

Derivative correctness Table 3 shows the step-sweep verification after the corrections: median relative errors fall from $\sim 10^{-6}$ at $\varepsilon = 1$ MVA to $\sim 10^{-10}$ at $\varepsilon = 10^{-3}$ MVA — $O(\varepsilon^2)$ convergence to the power-flow-tolerance floor — with adjoint residuals at machine precision. The non-convergent 10^{-2} -plateau of the original submission is fully explained by the two implementation defects above; no unexplained discrepancy remains.

DC–AC divergence Table 4 regenerates the paired radii with the corrected implementation. DC radii are bit-identical to the original table, as implied by slack invariance; AC radii shift moderately where the defects were active (case73: $14.24 \rightarrow 11.64$; case118: $6.18 \rightarrow 5.80$). The qualitative conclusion is unchanged: AC and DC radii differ materially (AC/DC from 0.45 to 1.16), so DC-only screening misjudges AC thermal robustness in both directions.

Table 5: Zero-flow line ends on `case2000_goc` (AC-FPF base point). Every nondifferentiable end now receives a first-order operator-norm radius, so the all-constraint certificate is defined; previously these ends were excluded and the case was only partially certified. End counts with $|S_0|$ below a threshold t (MVA): $\leq 1e-12$: 42, $\leq 1e-09$: 46, $\leq 1e-06$: 46, ≤ 0.001 : 46, ≤ 0.1 : 46 — the count is insensitive to the threshold over eight orders of magnitude, supporting the scale-aware default.

Quantity	Value
Monitored lines	2743
Lines with a zero-flow end	17
Operator-norm certified	17
All-constraint radius defined	yes
Global minimum radius	14.07
Q-limit-saturated buses absorbed as PQ	19
Max adjoint residual	2.29×10^{-15}

Table 6: Line-wise sigma calibration on the IEEE 118-bus system (load-proportional σ_P , $\sigma_Q = \sigma_P \tan(\arccos 0.9)$, 3000 nonlinear Monte Carlo replays, seed 42, 0 PF failures). Ten highest-variance binding ends. The analytical first-order standard deviation now matches the empirical nonlinear standard deviation (median ratio 1.004, range [0.989, 1.017]); the tightened-limit exceedance probability at $s_0 + 2\hat{\sigma}$ is compared with the empirical frequency and its Wilson 95% interval.

Line	End	$ S_0 $	Ana. sd	Emp. sd	Ratio	P_{pred}	P_{emp}	Wilson 95%
line_96	to	402.44	21.805	21.619	1.009	0.0237	0.0193	[0.0150, 0.0249]
line_90	from	244.62	15.204	15.070	1.009	0.0237	0.0223	[0.0176, 0.0283]
line_89	to	363.68	11.804	11.767	1.003	0.0231	0.0253	[0.0203, 0.0316]
line_88	to	169.21	10.805	10.731	1.007	0.0235	0.0200	[0.0156, 0.0257]
line_117	from	113.85	9.822	9.819	1.000	0.0228	0.0343	[0.0284, 0.0415]
line_49	from	134.73	9.433	9.272	1.017	0.0247	0.0270	[0.0218, 0.0334]
line_118	from	89.21	7.447	7.418	1.004	0.0232	0.0250	[0.0200, 0.0312]
line_128	to	92.94	7.284	7.274	1.001	0.0229	0.0200	[0.0156, 0.0257]
line_166	to	68.60	6.777	6.852	0.989	0.0216	0.0217	[0.0170, 0.0275]
line_110	from	263.10	6.542	6.605	0.991	0.0217	0.0250	[0.0200, 0.0312]

Complete certification of zero-flow ends Table 5 demonstrates the operator-norm extension on `case2000_goc`: all 17 lines with zero-flow ends now carry first-order radii, the all-constraint system radius is defined, and the nondifferentiable-end count is insensitive to the detection threshold over eight orders of magnitude.

Sigma calibration Table 6 reports line-wise calibration on the IEEE 118-bus system: the analytical first-order standard deviation matches the nonlinear empirical one with median ratio 1.004 (range 0.989–1.017) over the ten highest-variance ends, and tightened-limit exceedance probabilities fall inside or near the Wilson intervals. The $2.8\times$ miscalibration reported in the original submission was an artifact of the slack-index defect, not a property of the method; with the corrected sensitivities the sigma radius behaves as a calibrated first-order index *in this tested regime*. We retain the scope limitation: system-level risk is not the minimum line-wise index, and calibration under strong nonlinearity, larger perturbation scales, or heterogeneous covariance families requires the broader protocol outlined in Section 7.

Nonlinear validity range of the affine radius Table 7 quantifies, per case, where nonlinear violation actually occurs along the certified worst direction: the interpolated crossing sits at 0.90–

Table 7: Multi-scale nonlinear replay of the balanced worst-case direction for the five tightest lines per case, at scales $\alpha \in \{0.25, 0.5, 0.75, 0.9, 1.0, 1.1, 1.25, 1.5\}$ of the affine radius. “Crossing α ” is the interpolated scale at which the TARGET line first violates its limit in the nonlinear replay; values below 1 quantify the mild nonconservatism of the affine radius along its own worst direction. All replays converged; no voltage-band violations occurred at $\alpha \leq 1$.

Case	Lines	Med. crossing α	Min	Max	Violated at $0.9r$	Q-lim. events (base)
case14_ieee	5	0.952	0.902	1.002	0/5	3
case30_ieee	5	0.992	0.938	0.995	0/5	4
case118_ieee	5	0.993	0.980	0.996	0/5	29
case200_activ	5	0.999	0.958	1.000	0/5	21

Table 8: Ranking statistics on the IEEE 118-bus system with scenario-level uncertainty quantification (7612/8000 converged Monte Carlo scenarios, $\sigma_P = \sigma_Q = 30$, seed 42). Confidence intervals are SCENARIO bootstrap (resampling Monte Carlo scenarios, 2000 replicates), which treats the correlated line outcomes correctly, unlike the line bootstrap of the original submission; replicates re-estimate the empirical frequencies, so the intervals are mildly attenuated relative to the full-sample point estimate. The paired scenario-bootstrap 95% interval for the Spearman difference is $\Delta\rho = 0.159$, CI $[0.109, 0.161]$ vs. loading and $\Delta\rho = 0.182$, CI $[0.135, 0.182]$ vs. headroom.

Score	ρ	95% CI (scenario)	Prec.@5	Prec.@10	Top-5 mean freq.
AC L2 danger score ($1/r$)	0.781	$[0.738, 0.774]$	0.60	1.00	0.878
Loading ratio	0.621	$[0.599, 0.640]$	1.00	1.00	0.991
Headroom danger score	0.599	$[0.574, 0.619]$	0.60	1.00	0.870

1.00 of the affine radius (median 0.95–0.999). The affine radius is therefore mildly *optimistic* — by at most 10% and typically under 5% on these cases — and a user who requires conservatism can de-rate the radius by the reported case-wise factor. All replays converged; no voltage-band violations occurred at or below the certified radius.

Ranking quality with scenario-level uncertainty Table 8 repeats the 118-bus ranking study with 8,000 attempted nonlinear scenarios and scenario-bootstrap intervals. The radius danger score reaches $\rho = 0.781$ against 0.621 (loading) and 0.599 (headroom); the paired differences $+0.159$ and $+0.182$ have bootstrap intervals bounded away from zero. The radius-based danger score dominates loading- and headroom-only scores, and the *paired* confidence intervals for the Spearman differences exclude zero, which the line bootstrap of the original submission could not legitimately claim. Precision at k and top- k mean overload frequencies quantify the operational value directly.

Realizable response models With uncertainty restricted to load buses and generators responding through headroom-proportional participation factors, the certified radii change by -1% to $+21\%$ relative to the blockwise-balanced convention on the five tightest 118-bus lines, every worst-case perturbation respects all generator limits, and nonlinear replay at the certified radius reaches $|S|/c_\ell = 1.001$ – 1.015 on the target line — the certificate is simultaneously realizable and tight under this response model. This supports the central modeling claim: the response convention is part of the certificate geometry, and changing it changes the certified number in an auditable way.

Computational profile Table 9 separates power-flow, operator-build, and certificate stages with repeated timings and measured peak memory. The certificate stage (adjoint solves + norms, includ-

Table 9: Separated, repeated timings (mean $\pm$ std over 7 repetitions) and measured peak resident memory. Stages: nonlinear AC PF; operator build (Jacobian assembly + sparse LU); certificate (chunked adjoint solves + balanced norms, including the adjoint-residual check).

Case	Buses	Lines	AC PF (s)	Operator (s)	Certificate (s)	Peak RSS (MB)	Stored h (MB)
<code>case14_ieee</code>	14	17	0.240 $\pm$ 0.029	0.018 $\pm$ 0.005	0.030 $\pm$ 0.004	323	0.01
<code>case30_ieee</code>	30	34	0.275 $\pm$ 0.021	0.036 $\pm$ 0.007	0.054 $\pm$ 0.005	323	0.03
<code>case118_ieee</code>	118	175	0.355 $\pm$ 0.021	0.129 $\pm$ 0.009	0.502 $\pm$ 0.045	327	0.59
<code>case200_activ</code>	200	179	0.489 $\pm$ 0.147	0.186 $\pm$ 0.016	0.479 $\pm$ 0.029	330	1.09
<code>case2000_goc</code>	lossless runpp base point does not converge (AC-FPF base required, cf. Table 5)						

ing residual checks) remains sub-second through 200 buses on the revision hardware; the lossless `runpp` base point does not converge for `case2000_goc`, which is why the zero-flow study uses the AC feasibility power flow — the accounting is reported rather than hidden.

7 Discussion

What the certificate is A post-dispatch screening and diagnostic layer: exact for the affine surrogate, locally valid for the nonlinear system, calibrated in the tested regime, and honest about where the nonlinear crossing sits relative to the certified radius (0.90–1.00 on the studied cases). Under isotropic uncertainty several classical scores are monotone transformations of the same standardized margin; the value of the radius family is that it stays well-defined and auditable when the geometry is *not* isotropic — heterogeneous covariance, response constraints, participation maps.

What it is not Not a nonlinear feasibility certificate (unlike convex inner approximations [10]), not a system-level probability of overload, and not a dispatch method. Radius-aware dispatch is deliberately out of scope: a defensible evaluation requires matched-cost baselines (loading-, headroom-, PTDF-, and chance-based tightening under equal cost or equal cut budgets) and a recognized preventive/corrective SCOPF comparator across multiple lossy systems; that is a separate study.

Inequality-constrained uncertainty sets Box and polyhedral restrictions (generator headroom, ramp bounds, load nonnegativity, renewable availability) leave the closed-form family: the support function becomes a convex program (an SOCP for ellipsoid-plus-box). The participation-factor study shows one practically relevant middle ground that stays closed-form; the general trade-off between closed form and realism is inherent and should be chosen per application.

Limitations The AC certificate is local to the linearization at the base point, including its reactive-limit active set; large perturbations can change the active set along the path even when the base set is modeled correctly (the replay protocol reports these events). The lossless series-only surrogate is a modeling choice enforced consistently across certificate and verification; extending the Jacobian to the full pi-model (charging, shunts, transformer magnetizing branches) removes the need for the lossless policy and is mechanical but not yet implemented. Calibration evidence covers the tested uncertainty scale and one network family; base infeasibility, PF non-convergence on stressed cases, and islanding contingencies are reported as explicit statuses rather than silently dropped.

8 Conclusions

This paper provides a response-constrained thermal-security robustness radius as a verified post-dispatch screening and diagnostic certificate: a constrained dual-norm formulation with exact affine-model maximality, slack invariance, and metric-sigma equivalence; a sparse two-ended AC adjoint implementation whose derivative is machine-verified; operator-norm radii that certify zero-flow ends and make the all-constraint radius well-defined on large cases; calibrated sigma diagnostics; quantified affine-to-nonlinear crossing ratios; statistically defensible ranking evidence; and a realizable participation-response study. The framework is positioned within the steady-state security-distance literature as its constrained-geometry, AC-adjoint extension, and the dispatch-side use of the radius is left to a dedicated future study.

Data and Code Availability

Source code, entry points, tests, configuration files, revision experiment scripts, and their JSON artifacts are available in the public repository <https://github.com/nemakgregor/power-stability-radius> (branch `revision-r1-screening`; the exact commit hash accompanies each artifact). An archived release with a DOI will be minted at resubmission. Benchmark cases are public PGLib-OPF inputs [14].

References

- [1] Allen J. Wood, Bruce F. Wollenberg, and Gerald B. Sheblé. *Power Generation, Operation, and Control*. Wiley, 3 edition, 2013. ISBN 9780471790556.
- [2] Dimitris Bertsimas and Melvyn Sim. The price of robustness. *Operations Research*, 52(1): 35–53, 2004. doi: 10.1287/opre.1030.0065.
- [3] Aharon Ben-Tal and Arkadi Nemirovski. Robust convex optimization. *Mathematics of Operations Research*, 23(4):769–805, 1998. doi: 10.1287/moor.23.4.769.
- [4] Daniel Bienstock, Michael Chertkov, and Sean Harnett. Chance-constrained optimal power flow: Risk-aware network control under uncertainty. *SIAM Review*, 56(3):461–495, 2014. doi: 10.1137/130910312.
- [5] Line Roald, Sidhant Misra, Michael Chertkov, Scott Backhaus, and Goran Andersson. Chance constrained optimal power flow with curtailment and reserves from wind power plants. *arXiv preprint arXiv:1601.04321*, 2016. URL <https://arxiv.org/abs/1601.04321>.
- [6] Miles Lubin, Scott Backhaus, and Yury Dvorkin. A robust approach to chance constrained optimal power flow with renewable generation. *IEEE Transactions on Power Systems*, 31(5): 3840–3849, 2016. doi: 10.1109/TPWRS.2015.2499753.
- [7] Sijie Chen, Qixin Chen, Qing Xia, and Chongqing Kang. Steady-state security assessment method based on distance to security region boundaries. *IET Generation, Transmission & Distribution*, 7(3):288–297, 2013. doi: 10.1049/iet-gtd.2012.0288.
- [8] Sijie Chen, Qixin Chen, and Qing Xia. Steady-state security distance: Concept, model and meaning. *Proceedings of the CSEE*, 35(3):600–608, 2015. doi: 10.13334/j.0258-8013.pcsee.2015.03.012.

- [9] Sijie Chen, Qixin Chen, Qing Xia, Haiwang Zhong, and Chongqing Kang. N-1 security assessment approach based on the steady-state security distance. *IET Generation, Transmission & Distribution*, 9(15):2419–2426, 2015. doi: 10.1049/iet-gtd.2015.0552.
- [10] Hung D. Nguyen, Krishnamurthy Dvijotham, and Konstantin Turitsyn. Constructing convex inner approximations of steady-state security regions. *IEEE Transactions on Power Systems*, 34(1):257–267, 2019. doi: 10.1109/TPWRS.2018.2868752.
- [11] Laurențiu Lucian Anton and Marija Ilić. Unified diagnostics for quantifying ac operating-point robustness under injection and topological uncertainties with regime changes. *International Journal of Electrical Power & Energy Systems*, 178:111979, 2026. doi: 10.1016/j.ijepes.2026.111979.
- [12] Dongchan Lee, Konstantin Turitsyn, Daniel K. Molzahn, and Line A. Roald. Robust ac optimal power flow with robust convex restriction. *IEEE Transactions on Power Systems*, 36(6):4953–4966, 2021. doi: 10.1109/TPWRS.2021.3075925.
- [13] Abraham M. Hasofer and Niels C. Lind. Exact and invariant second-moment code format. *Journal of the Engineering Mechanics Division*, 100(1):111–121, 1974. doi: 10.1061/JMCEA3.0001848.
- [14] Sogol Babaeinejadsarookolaee, Adam Birchfield, Richard D. Christie, et al. The power grid library for benchmarking ac optimal power flow algorithms. arXiv:1908.02788, 2021. IEEE PES PGLib-OPF Task Force.
- [15] Ray D. Zimmerman, Carlos E. Murillo-Sánchez, and Robert J. Thomas. Matpower: Steady-state operations, planning, and analysis tools for power systems research and education. *IEEE Transactions on Power Systems*, 26(1):12–19, 2011. doi: 10.1109/TPWRS.2010.2051168.
- [16] Leon Thurner, Alexander Scheidler, Florian Schäfer, Jan-Hendrik Menke, Julian Dollichon, Friederike Meier, Steffen Meinecke, and Martin Braun. Pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. *IEEE Transactions on Power Systems*, 33(6):6510–6521, 2018. doi: 10.1109/TPWRS.2018.2829021.
- [17] Alinson S. Xavier, Aleksandr M. Kazachkov, Oğuzhan Yurdakul, Jun He, and Feng Qiu. Unit-commitment.jl: A julia/jump optimization package for security-constrained unit commitment, version 0.4. Zenodo, 2024.